Jun 16, 2025

Stephanie Carlton, Deputy Administrator of the Centers for Medicare and Medicaid Services
Steven Posnack, Acting Assistant Secretary for Technology Policy
Centers for Medicare and Medicaid Services
Department of Health and Human Services
Attention: CMS-0042-NC
P.O. Box 8013
Baltimore, MD 21244-8013

RE: Request for Information; Health Technology Ecosystem

Dear Ms. Carlton and Mr. Posnack,

On behalf of the knowledgeable and innovative team of technologists here at Quadratic Digital, we appreciate the opportunity to provide comment on the Centers for Medicare and Medicaid Services (CMS) request for information (RFI) regarding America's Health Technology Ecosystem. As we look to the future, it is only through collaboration, courage, and conviction that we are able to forge decisive solutions resulting in a better tomorrow for all Americans. It is abundantly refreshing to see that CMS is seeking poignant, directed feedback from Patients, Payers, and Vendors alike, as this most assuredly will provide the most cohesive "big-picture" possible.

Quadratic Digital believes that CMS is uniquely positioned to effectuate very meaningful outcomes, not only for patients, but for all stakeholders in American healthcare infrastructure, through the implementation of meticulously crafted solutions leveraging a wide range of novel technologies. As America's guiding health agency, CMS has a long history of safeguarding and enabling the American people to have improved access to valuable healthcare and coverage, dating back to its origination as the Healthcare Financing Administration (HCFA). The advent of Value-Based care models and Alternative Payment models speaks volumes to the inextricable coupling of healthcare's delivery and financial implications.

Quadratic Digital strongly believes that critical aspects for a comprehensive application of technology resulting in better outcomes for patients and attainable, affordable health coverage include improved interoperability between stakeholders, meticulous use of cloud computing capabilities, and targeted use of interpretive analytics leveraging machine learning. We have attached our responses to your most critical inquiries.

As trusted technology partners to CMS, and foremost, as Americans, the team at Quadratic Digital recognizes a civic duty to aid in accomplishing these goals for our fellow Americans and want to thank CMS for the diligent dedication to this mission. As you well know, America's Health Ecosystem is vastly complex, with a wide array of stakeholders and operations in play; each with a crucial role to health outcomes for the American people. Together, in unity, I am confident that these outcomes can be improved to fortify the American Health System through

data-driven decisions and observation of outcomes. I hope you find our insights to be beneficial as the agency plans its roadmap to success. If you have any questions regarding the team's or would like to discuss further, please feel free to reach out to me at Andy.Parr@quadratic-digital.com.

Sincerely,

Andy Parr
Chief Executive Officer
Quadratic Digital, LLC

Responses to CMS' Request for Information; Health Technology Ecosystem

June 16, 2025

Foreword

The responses contained within represent the perspectives of multiple tenured technologists with a wide range of knowledge within the healthcare and technology sectors. The team appreciates the opportunity to provide insight on the path forward in America's healthcare journey, and is optimistic that their contributions will provide a positive impact for fellow Americans, stakeholders, and the United States Government alike. The inquiries found within Section E of the CMS Request for Information identified as "CMS-0042-NC" are restated below, with the correlating recommendation from the Quadratic Digital Response Team.

Response Team Qualifications

- Vast knowledge of Medicare Policy, Cloud Computing, and Legacy Federal Systems
- Over 15 Years Experience in:
    - Coordinating Software Delivery on various CMS Programs
    - Software Engineering on various CMS & Federal Programs
    - Cloud and On-Premises Infrastructure
    - Cloud Solutions Design
- Proficiencies:
    - Anthropocentric Design & Analysis
    - User Experience Design
    - Real-time and Batch Data Processing
    - Enterprise, Service-Oriented, and Microservice Architecture
    - Full-Stack Applications & Web Development
    - Legacy Infrastructure Modernization
    - Legacy Applications Modernization

Inquiry Responses

1. Ecosystem

Inquiry TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

TD-1 Response: The complexity of this inquiry cannot be understated. There are a number of excellent opportunities which exist, both short and long term, for Application Development Organizations to create digital health products for Medicare beneficiaries and caregivers. This development consumes resources quickly and can often prove costly to get off the ground. This can prove prohibitive to many of technology's best assets. CMS has long adopted contract vehicles and closely-monitored programs and projects to implement software solutions closely aligned with Medicare's needs, however, *true* grassroots product development differs from

service-oriented delivery. While the existing acquisition process is competitive and outcome oriented, a Grant-style approach, with key outcomes defined as a condition of accepting the Grant, may afford development organizations the flexibility and freedom needed, while retaining a degree of vigilant oversight on output. As a provision of Grant acceptance, organizations would agree to provide the end product to CMS free of charge. This also negates any benefit to developing a product that would require extensive ongoing maintenance or prolonging a contract period, which, while unfavorable to consider, has likely played a role in prior procurements.

Additionally, Medicare has a particularly nuanced implementation to healthcare (granted the unique use case presented, as a public entitlement), especially from the payer perspective. Although confidentiality of processing structure is advantageous to preventing fraud and misuse, in consideration of this nuance, grassroots product development should be afforded the necessary information on Medicare Processing to build to Medicare's spec - which would likely cover many commercial use cases, and allow grassroots products to sustain from a potential parallel commercial revenue stream. Far too often we have seen dedicated development projects for Medicare's specific use case which could be abstracted and applied in many contexts, if the parameters and context of their development were adapted to support commercial and private sector growth as well.

Inquiry TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

TD-2 (a) Response: CMS may find that it is advantageous to provide detailed, deidentified claims data to research partners, as well as customizable, aggregated claims data, enabling partners to conduct through analyses on specific services, diagnoses, and geographic regions to seek out patterns and commonalities between the data points. For instance, if a data analytics partner sought to find which diagnoses were most common amongst the Medicare population, or which service proved most costly in a particular year, these features would be most valuable - and could lead to significant scientific development, which ultimately could reduce the observed impact and provide real, tangible results to the Medicare patient population.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

TD-2 (b) Response: The Blue Button 2.0 API surfaces a wealth of data as it exists today, and its implementation using Fast Healthcare Interoperability Resource (FHIR) Implementation Guides create a cohesive and unified approach to retrieving data. Despite this, an interim layer is necessary for many Medicare Beneficiaries to utilize the data as it is surfaced today, and the existence of such interim layers is not as widespread. In order for beneficiaries to make better informed decisions about their healthcare, they must be enabled to be aware of the data surrounding it, which Blue Button enables rather well. This speaks volume to the importance of how the data is presented in the context of its audience as well.

While the data available from Blue Button is vast and comprehensive from a Medicare perspective, from a patient perspective, access to crucial health information such as lab results, imaging, vital signs, and other determining factors of health would greatly assist beneficiaries and stakeholders in understanding their health in a comprehensive, holistic approach. With this information, healthcare providers and patients become better equipped to live healthier lives, reducing the need for interventional treatments and ultimately experiencing better outcomes and lower program costs through prevention.

A unique, centralized repository for this data, surfaced in a deidentified manner could also prove immeasurably valuable on a research front as well, giving analysts and scientists closely regulated access to huge amounts of clinical data and enabling thorough analysis and identification of causal relationships and trending in American healthcare.

c. What obstacles prevent accessing these data sources today?

TD-2 (c) Response: Although a vast array of Health Information Exchanges are available for exchanging healthcare data between entities, the challenge persists in establishing uniformity and sourcing that enable all individuals to leverage a single access point for information. Multiple permutations of data models, data availability, and provider engagement increase development complexity and often Application Development Organizations find themselves working through a backlog of tasks geared toward expanding data availability, rather than true feature development. This work proves cost prohibitive and ultimately does not lead to successful products in the name of supporting a sufficient number of benefactors. Having a sole source for health data would open many pathways and aid in the success of future endeavors. Additionally, alongside FHIR, a clear set of standards to be leveraged enterprise or industry-wide, such as Javascript Object Notation (JSON) API contracts, as well as Comma-Seperated Values (CSV) and Database Schema Naming conventions would enable efficient development of a small set of utilities supporting interoperability through conversion and translation between data entities.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

TD-2 (d) Response: In addition to continuing support of Blue Button 2.0 and other internal APIs, CMS should consider an API centered around Pricing Transparency which allows Medicare Beneficiaries to accurately assess their costs relating to upcoming procedures and appointments in advance in a holistic manner.

Because Medicare is a public entitlement, heavily guided by mandates and statutes, accurate pricing or even estimation of pricing relies on a multitude of information available within different components of the CMS Ecosystem, such as CWF for utilization data, PECOS for provider status, and of course, PCS for the pricing calculations and fee schedule data itself.

Leveraging the Advanced Explanation of Benefits bundle from FHIR, CMS could build this API in a manner that allows for integration with many Electronic Health Record (EHR) systems, and

easily integrate it into the MyMedicare Beneficiary Portal, addressing many use cases with a single solution that, with proper integrations, prove highly valuable to patients and providers alike.

CMS should also consider including a robust API capability requirement in any future PECOS redesign or modernization, to properly surface Provider Enrollment Information and status in a real-time capability that offers greater reliability and availability. The CMS 855 process could become entirely API driven and also integrated directly into EHRs, streamlining the enrollment of new providers for healthcare organizations across the country.

Finally, an API surfacing de-identified Medicare claims and quality data would provide great transparency for research organizations and healthcare providers alike to identify trends and gaps as they occur in America's Health System. As CMS makes strides toward processing claims more efficiently and quickly, this data can be turned around to forecast potential issues and prioritize research efforts for immediate and effective value.

2. Digital Identity

Inquiry TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

TD-3 (a) Response: Protected Health Information (PHI) and Personally Identifiable Information (PII) are of the utmost importance when considering the nature of Healthcare data, and must be protected at all times. The flagship method for ensuring this protection is through the use of digital identity services to authoritatively assert an individual's identity before granting access to PII and PHI. With a robust digital identity service, CMS and ONC alike could validate an individual's identity as well as, or even more effectively than a physical exchange of credentials. A well designed and implemented solution would provide a cohesive and convenient access solution to Federal resources, improving user experience and reassuring trust in the government and applications, while CMS and ONC would have a more robust and less vulnerable fraud posture.

A primary challenge with digital identity is during the issuance of a credential - if a credential is falsely issued to the incorrect entity, any activity downstream of issuance with that credential is equally as detrimental. Many digital identity solutions have turned to validation steps using facial recognition software and vetting identification against government issued IDs. Multi-factor Authentication (MFA) has also become increasingly widespread in use for digital identity verification, requiring users to have multiple vetted access points (such as accounts, devices, telephones, etc.) before granting access and confirming an identity. This can be cumbersome due to the possibility of lost devices and unrecoverable accounts, both of which should require invalidation of the credential and subsequent update.

Once issued, management of credentials to prevent compromisation and more importantly, revoking compromised or potentially compromised credentials is critical to system security. With the nature of CMS' mission often relying on patients having proxy representatives, this is a

crucial capability that should not be overlooked, with proxy capability optimally transcending the digital plane so that representatives can be designated on a moment's notice if necessary, even if in a hospital or provider's office.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

TD-3 (b) Response: Requiring digital identity credentials would better align CMS and ONC with other Federal agencies on policies relating to access to Federal systems and greatly enhance the cybersecurity posture of the agency, providing heightened reassurance for protection of personal and non-personal information. This posture comes at a trade-off, however, as data exchange itself would inherently become more meticulously regulated and require additional tasks and maintenance to establish connectivity and authorization for the impacted systems. Additionally, with multiple credentialing providers available for the service, CMS faces the challenging question of which one, or more specifically, which one(s) to leverage. Multiple options can help keep data exchange open and available, while simultaneously introducing additional technical burden for maintaining multiple integrations. The CMS Portal and CMS Identity Management (IDM) systems provide an excellent vehicle for introducing these credentialing services to the greater CMS Enterprise while regulating the technical burden these services would introduce. CMS' Enterprise User Administration (EUA) leverages a detailed capability with Job Codes enabling organizational unit to manage access to their own applications, and this capability should be extended upon.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

TD-3 (c) Response: OpenID Connect (OIDC) provides a highly effective method for authorization and exchange of user profile information, and is regarded as the de-facto industry standard for this use case. Mandatory use of OIDC, however, in the context of the CMS Enterprise, would likely incur an influx of required development to update many existing enterprise services as Secure Assertion Markup Language (SAML) 2.0 still has a significant footprint in the industry and is a direct alternative to OIDC still in use by key identity providers today. A progressive migration of existing applications to OIDC, while requiring new applications to launch with the protocol, would unify CMS and ONC's posture on identity protocol and foster a more cohesive digital identity stance for the agencies.

3. Technical Standards and Certification

Inquiry TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

TD-4 Response: Standards-based APIs, such as Representational State Transfer (REST), are highly customizable for a vast array of use-cases which allow greater flexibility in development, maintenance, and operation of systems. CMS has several successful implementations of these in their technical portfolio. To encourage interoperability and further use of open standards-based APIs, CMS should consider including this as a requirement in upcoming

procurements to ensure the agency accurately establishes expectations upfront for the outcome of development projects. Proprietary APIs, while merited for certain use cases, often become very nuanced and specific implementations that require a wealth of context and knowledge of subject matter to adequately maintain, leaving CMS with little to no choice when approaching maintenance re-tooling options. In summary, CMS should seek to reduce the fringe incentive of on-going work that is established by developing nuanced, highly proprietary solutions, and instead, incentivize use of public tooling wherever possible. Given the centric position of CMS, as the agency continues to standardize and define the modern approaches to be leveraged, it will become easier for contractors and government agencies alike to reduce the siloed nature of data exchange.

Inquiry TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

TD-5 Response: Establishing what is essentially the base Domain Name Server (DNS) for accessing FHIR data will greatly reduce the barriers for patients to access their data. These are the users who will have the most difficulty at present, and focusing on the patient experience will maximize impact and value. Those listed in the directory should bear at least a portion of the cost and would prove valuable by eliminating the need for providers to provide a “front page” for their FHIR API (and/or avoid paying for a commercial product). This, in essence, would provide a model similar to that utilized by cloud service providers, where cost is assessed for actual usage and the volume of data exchanged, rather than by flat-rate or other method.

Inquiry TD-6. What unique interoperability functions does TEFCA perform?

TD-6 Response: TEFCA, at its roots, provides a unified agreement between Qualified Health Information Networks (QHINs), which creates a common-ground for the establishment and implementation of interoperability in America. Without such an agreement, operation between multiple entities can, and oftentimes, does become counter-intuitive, with many teams spending countless hours building mechanisms to adapt to each other’s design approaches and interfaces. The inception of TEFCA introduced a centralized aspect of governance, cohesion, and uniformity that was previously absent from interoperability, which would be challenging to replicate without such an entity.

a. What existing alternatives should be considered?

TD-6 (a) Response: Although several alternatives to TEFCA exist, many result in only partial success; a patient-centric approach to data interoperability is the only existing alternative that could realize substantial benefit, of which TEFCA serves as an enabling component. Allowing patients to retrieve their own data, and disseminate it appropriately, aligns with American values, but also has the drawback of patient burden, which, in the context of the healthcare setting, is often not an ideal implementation. Human-centered research has taught us that healthcare data is always critical, but especially so in the acute clinical setting, where time is of the essence, and the patient is not always available due to injuries or illness. In such a situation, it is especially

important that providers have the capability to retrieve a patient's healthcare data without patient or representative action, to enable the most positive outcomes.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

TD-6 (b) Response: Because TEFCA was implemented in a phased approach, there are inherently redundant standards in the legacy formats, such as Simple Object Access Protocol (SOAP), which would not be leveraged in a modern implementation. Although this redundancy exists, a crucial pillar to the success of TEFCA and interoperability is the continuous, uninterrupted flow of data and the ability for consumers to integrate at an asynchronous pace, to avoid disruptive development through strict implementation timeframes. This means the current redundancy observed actually serves to bridge the gap between the old and the new in the context of American Healthcare Information Technology, however the importance of establishing a cutover timeframe and deprecating the legacy standards of TEFCA cannot be understated.

Inquiry TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

TD-7 Response: USCDI has established a unified collection of data for interoperability between entities and networks in the healthcare ecosystem. Much like TEFCA, such a collective agreement or baseline of the "rules of engagement" is crucial to ensuring that a distributed network operates cohesively and in synergy. Unfortunately, due to the volume of entities who must implement USCDI, ongoing updates to reassess and enhance the data posture is challenging beyond the initial release without closely monitoring implementation and carefully planning how these updates are disseminated.

a. Does it contain the full extent of data elements you need?

TD-7 (a) Response: Our team has not yet had a requirement involving USCDI, and therefore cannot opine on the comprehensive coverage of the data.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

TD-7 (b) Response: Not applicable.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

TD-7 (c) Response: Not applicable.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

TD-7 (d) Response: One of the key advantages of a uniform data interoperability model is the promotion of data consistency and processing flexibility. In the hardcopy world, paper forms establish a standard of communication to allow an organization to collect data from many different entities without tailoring the approach to each entity. Further reducing structural format requirements may introduce inconsistencies in data, scale the receiver's burden horizontally (need to accommodate multiple implementations) and the use of language models is not yet sufficiently reliable to provide a repeatable expected outcome with assurance. Large Language models (LLMs) will be very useful in transforming historical data into modern formats, although removing structure could increase analytical error rate due to inconsistencies. LLMs will also need to adjust to changes in terminology over time and are particularly susceptible to jargon, as they may struggle to identify instances where two different terms come to carry the same intended meaning from humanistics.

Inquiry TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

TD-8 Response: The ONC certification criteria for Patient Demographics and Computerized Provider Order Entry have proven to be critical assets to a well-postured Electronic Health Record, as they enable tracking of critical identification information as well as a means for tracking and transmitting necessary activity to improve a patient's health.

Inquiry TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

TD-9 (a) Response: Moving to prioritize API-enabled capabilities would be similar to raising the bar of standards for EHRs. Gainful software functionality supporting the API transactions is an inherent requirement when requiring the ability to support the exchange of data in an API format. This could be surfaced as a tiered certification approach, offering different levels of certification depending on the maturity level of the software under development. The highest level would define the vision for an ideal implementation and include robust API capability, while still providing developers with a meaningful vehicle for iterating and progressively reaching the target.

b. What would be the drawbacks?

TD-9 (b) Response: As seen with many software projects through the years, redirecting requirements can be risky as although inheritance would be present (data cannot be available through an API if it is not stored or received), oftentimes requirements that are not explicitly stated are overlooked or omitted in the interest of expedited delivery. Functionality and interoperability should be given equal consideration, as both are crucial to meaningful use of EHR products.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

TD-9 (c) Response: The primary mechanism for ONC to support exchanging patient data would be to identify a recommended format or convention for the exchange of data, thereby enabling all industry partners to work around a unified expectation for interfacing - or a bundle of formats, with a utility for interchange.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

TD-9 (d) Response: As seen with the EHR Incentive Program, providers respond well to incentive. Given that the National Level Repository has long been established and is still in operation today, this system could provide an existing mechanism for further incentive-based programs to offset the cost of technical adoption of new direction and encourage providers to implement new directives. CMS could also seek to provide open-source implementation libraries that would enable all partners to quickly and easily integrate with a unified approach on a level playing field.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

TD-9 (e) Response: Quality data is most beneficial when all cases are assessed in a holistic fashion, otherwise the data skews. Given the sheer breadth of the American population, any expectation to transmit or provide this data in a non-bulk manner would require continuous transmission and likely a high level of effort for providers and third-parties. Again, a unified format for quality data is a key pillar to implementing bulk transfer in an efficient and meaningful manner, however CMS should also seek to manage processing of this data in an effective manner as well. Allowing EHRs and providers to transmit large amounts of data to a landing, or staging zone, which is well-secured, would allow CMS to process storing of the data asynchronously and alleviate much of the burden during transmission. This could easily be accomplished using simple block storage and lambdas. CMS could also encourage EHRs to perform adequate de-identification of the data before transmission, which would reduce administrative burdens on CMS, and improve the privacy posture of the transmission process itself.

Inquiry TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

TD-10 Response: ONC's best option for implementation of the API condition within the Cures Act is to provide a formal recommendation, or playbook, on how implementation should be executed. This greatly reduces the burden for implementation teams on the EHR and provider side as it would serve to reduce a repetitive analysis and enable teams to move to actual development faster.

Inquiry TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

TD-11 (a) Response: The late opinion would provide challenges to some organizations as they adapt to conform to standardization requirements, however the value of issuing a standardized specification will likely be realized for years to come. This capability should undergo a revision.

b. Are there specific workflow aspects that could be improved?

TD-11 (b) Response: The capability should be evaluated holistically, with a focus on introducing asynchronous communication to reduce technical processing burdens.

c. Should CMS consider policy changes to support this capability's use?

TD-11 (c) Response: CMS should continuously consider policy changes in support of novel capabilities to encourage adaptation and implementation which enhances the posture of America's Healthcare IT Infrastructure.

4. Data Exchange

Inquiry TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

TD-12 Response: CMS should only consider endorsing non-CMS data sources and networks once a thorough analysis of the data and architecture can establish that the entity is reliable and secure. Transmission rates, security, scalability, and operational stability or redundancy should be regarded as critical for any organization's assessment.

Inquiry TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

TD-13 Response: This casts a wide net of opportunity from research to trend analytics, which could provide the insight necessary to begin more accurately forecasting healthcare issues in America and recognizing likely outcomes sooner, which may improve prognosis. Americans, armed with information, would likely be able to make more educated decisions about their healthcare and be better poised to address health concerns though prevention, rather than treatment.

a. What are the primary obstacles to this?

TD-13 (a) Response: Privacy, availability, and scalability, are hallmark challenges. It is also of note, the Medicare demographic is typically less technologically adept and able, however this may shift as generations cycle through the program.

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

TD-13 (b) Response: No remarks.

Inquiry TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

TD-14 (a) Response: Not applicable.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

TD-14 (b) Response: Not applicable.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

TD-14 (c) Response: Not applicable.

Inquiry TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

TD-15 (a) Response: Bulk FHIR provides the advantage of a unified data model that enables all parties the ability to have equal expectations of the data it is exchanging. As bulk data transmission inevitably grows, this unified format would circumvent the burden of transforming different data formats to the preferred format of the developer.

b. What are the potential disadvantages of their use?

TD-15 (b) Response: Scalability is more challenging with bulk data transfer, as all points in the data flow must account for the larger data transmission.

Inquiry TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

TD-16 Response: No remarks..

a. Do current rules encourage scalable network participation?

TD-16 (a) Response: No remarks.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

TD-16 (b) Response: No remarks.

Inquiry TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

TD-17 Response: No remarks.

5. Compliance

Inquiry TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

TD-18 (a) Response: Information blocking has often appeared, in our experience, as more of an oversight during architecture or design, rather than a deliberate practice. Improper infrastructure configuration, such as timeouts, or dropped connections are particularly troublesome as the data becomes effectively unusable.

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

TD-18 (b) Response: As stated above, the practice of information blocking has presented as more of a gap in design or development, rather than deliberate action. Accordingly, if CMS and ONC were to establish a baseline Service Level Agreement (SLA) with quantifiable metrics on availability and performance that must be met through scaled performance testing as a provision of certification, this could be addressed.

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

TD-18 (c) Response: No remarks.

Inquiry TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

TD-19 (a) Response: Given the nuanced nature of health insurance and the multitude of factors surrounding a health insurance claim, indications suggest that the biggest shortcoming of pricing transparency is not the ability to simply understand a price, but rather the ability to preemptively determine if a service or procedure will in-fact, be covered at all. This has a significant impact on the end price a patient will pay, and often depends on factors that are much more complex than simply looking up a rate. Without this component, a significant gap persists in patients being accurately informed on the financial implications of their healthcare.

b. Which workflows would benefit most from functional price transparency?

TD-19 (b) Response: Pricing transparency is beneficial in multiple workflows, beginning with the treatment decision process and extending through claims adjudication and into budgetary considerations for public healthcare programs such as Medicare and Medicaid. As a people, we cannot simply assume all persons are free from a financial restriction to getting the health services they need to thrive, and should seek to provide sufficient accurate information to support those decisions.

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

TD-19 (c) Response: Patients and providers would benefit greatly from an Advanced Explanation of Benefits (AEOB) implementation, allowing both entities to essentially “mock” the processing of a claim, and have an earlier understanding of the fiscal outcome relating to different treatment outcomes.

From a payer perspective, these AEOBs could potentially be converted to prior-authorizations, expediting the claims adjudication process and enabling better forecasting of expenditures for non-acute illnesses and procedures.

d. What would further motivate solution development?

TD-19 (d) Response: The various contexts in which pricing transparency should be implemented results in multiple avenues of motivation. With Medicare, CMS is the lone entity with sufficient view into how a Medicare claim is processed to implement a meaningful pricing transparency solution. Current offerings through MACs and public use files is beneficial, but not

sufficient for Medicare patients and representatives to accurately assess the financial implications of a service or procedure, especially in the context of Medicare utilization rules.

For the private sector, payers are most certainly the entity upon which this burden falls. A grant program, in which payers may recoup a portion of the implementation cost for a meaningful pricing transparency program, which is validated against a core set of functionalities and capabilities would help to encourage private insurers to adopt pricing transparency as a priority and bring better healthcare to millions of Americans.